

Hierarchical Federated Multi-Cluster Fog Architecture with Dynamic Node Allocation for Massive Scale IoT Deployments

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Abstract—Today’s systems of fog computing are having problems in supporting million-device IoT deployments because of single points of failure and fixed resource allocation [1]. In this paper a Hierarchical Federated Multi-Cluster Fog Architecture (HFMCFa) that uses flexible node allocation and a system of fault-tolerance with a master-of-masters coordination is presented. The architecture has four main layers: IoT devices, fog nodes with gateways at the level of clusters, regional masters, and global master-of-masters layer that includes a backup system (master-slave) for trustworthiness [2]. Each cluster starts with one fog node, one master node and 3-7 sleeping nodes that can turn on instantly based on the fact that the system predicts load algorithms. Each cluster’s size changes when the models use request frequency (R), user density (U), processing capacity (C), and latency constraints (L). The master-slave architecture prevents service failure automatically when one master stops working [3]. According to tests, the system can support 1,000,000+ IoT devices with 99.95% availability, 67% faster response duration compared to cloud-centric approaches, 45% better resource use and failure recovery in about 2.3- seconds. When we compare with normal single-cluster models, it shows 4x better scalability and it also keeps high energy efficiency due to its smart management of the sleeping node [4]. The introduced architecture offers a reliable solution for critical challenges in smart cities, industrial IoT and automated systems while requiring large-scale edge computing with high reliability and strong fault tolerance [5].

Keywords—Hierarchical Fog Computing, Dynamic Node Allocation, Master-of-Masters, Fault Tolerance, IoT Scalability, Load Balancing

I. INTRODUCTION(HEADING1)

IoT deployments are having challenges that have not been experienced before. By 2030, more than 125 billion devices are expected to generate an enormous amount of data [6]. Moreover, fog computing systems can’t handle metropolitan-scale deployments easily due to several factors:(1) they rely on single points of failure at federation levels[7], (2) they use fixed resource allocation which limits efficiency [8], (3) they lack a hierarchical coordination for large deployments[9],

and (4) they don’t have smart energy management methods for sleeping nodes [10].

In our previous research adaptive fog computing systems for IoT devices that supported mobility and location awareness were demonstrated [11]. However, when the number of connected devices exceeded 100,000, the systems ran into major scalability challenges [12]. Recent studies have revealed the need for master-slave architectures in systems to achieve fault tolerance and high availability results [2,13].

In this paper HFMCFa with several key contributions are presented: (1) fault-tolerant master-of-masters system that prevents single points of failure using hierarchical master-slave backups, (2) dynamic node allocation algorithms that assign tasks to nodes efficiently based on real-time load prediction,(3) smart management of sleeping node that save 38% energy while keeping QoS guarantees, and (4) strong scalability that support 1,000,000+ devices with 99.95% availability through distributed fault tolerance methods[14].

II. EASE OF USE

A. Related Work

Progressions in fog computing research have made significant advancements, particularly in the areas of hierarchical architectures and fault tolerance features [1]. The latest reviews show that load balancing techniques in fog computing are commonly used and that three-layer (IoT-Fog-Cloud) architectures are still predominant [15].Moreover, current methods don’t fully use master-of-masters methods for large deployments. Mohammadi et al. [3] have investigated researching fault tolerance in fog-based social IoT and investigating decentralized systems as a means of better resource allocation in fog computing. While some methods of conserving energy through wireless sensor network technologies exist within the fog computing context, there appears to be no existing methods for managing sleeping nodes, a key component in sensor networks[16].

Industrial IoT Applications require fog systems with mobility-aware design, and this can best be achieved by installing fog nodes according to a hierarchical approach [17]. In the past few years many studies focusing on IoT Device Development have shown that by combining fog computing

with other IoT technologies, you will see improvements in both energy efficiency and latency when you compare it to using only Cloud Technology [18]. Recent investigations have mainly been conducted using only two-tiered hierarchy approaches; however, a multi-tier hierarchy approach is necessary for million-device IoT deployments. Furthermore, our previous work [11] showed how to implement a mobility aware fog computing architecture but did not include features like hierarchical fault-tolerance needed for reliable operation on a large scale.

III. HIERARCHICAL FEDERATED ARCHITECTURE

A. Four-Layer Hierarchical Structure

A hierarchical structure is used in the system HFMCFCA to solve the issues such as scaling and keeping the reliance of the large IoT deployments. Figure 1 shows the architecture consisting of four layers: (1) The IoT Device Layer includes devices with limited resources such as sensors, actuators, and any type of mobile device capable of only basic methods of communication. (2) The Cluster Layer contains gateway multiplexers, worker fog nodes, sleeping nodes, and cluster masters capable of managing all aspects of the local configuration. (3) The Regional Layer contains regional masters, which organize multiple fog clusters within certain geographic areas. (4) The Global Layer is home to the master-of-masters node (with redundancy in master/slave mode), which manages all areas of the Global Management System and has a fault-tolerant design.

TABLE I. HFMCFCA SYSTEM CONFIGURATION

Tier	Component	Initial Config	Function
Global	Master-of-Masters	1 Master +2-3 Slaves	Global coordination, fault tolerance
Regional	Regional Masters	5-10 per region	Inter-cluster coordination
Cluster	FogNodes+Master	1 Active +3-7 Sleeping	Local processing, Load balancing
IoT	Devices	1000-5000 per cluster	Data generation, sensing

B. Master-of-Masters Coordination

Due to its adoption of the Master-of-masters deployment model, a master-slave model, instant backup activation can happen and the system will function appropriately even if issues exist. The Global Master Node oversees the entire deployment. It establishes overall rules of the application and provides methods for communication with the worldwide application. Slave Master Nodes can become backups in sync and they can be changed to masters fast, in 2.3 seconds, in case of failure. The State Synchronization Protocol uses Raft (an improved method) which can easily fit to conditions of network [19].

C. Dynamic Cluster Initialization and Sleeping Node Management

A gateway multiplexer in a cluster directs requests using the energy-efficient scaling properties of that multiplexer. Each cluster also contains one worker fog node that processes the cluster's main tasks and one cluster master that oversees

the local activities of all nodes and 3-7 sleeping nodes that conserve energy. Active sleeping nodes use an average of 15% of their active power [16]. Algorithms predictive of previous actions and current needs will allow sleeping nodes to activate in a duration of time between 30 to 60 seconds [20]. While this arrangement lowers the amount of energy consumed, it maintains maximum service quality through the effective management of resources.

ALGORITHMS FOR DYNAMIC NODE ALLOCATION

A. Mathematical Model for Determining Optimal Cluster Size

N_{opt} = the optimal cluster size

We can use organized mathematical models to find the optimal cluster size by checking how the nodes work under different kinds of workloads. The optimization formula helps in figuring out of the best setup by testing different options:

$$N_{opt} = f(R, U, C, L, Q_{target}, E_{budget})$$

R = request frequency (req/sec), U = user density, C = workload capacity, L = limitation of latency, Q_{target} = QoS boundary,

E_{budget} = cut off of energy budget.

The full optimization considers factors like load balancing η (0.7-0.9), possible system failure $P_{failure}$ (0.05), and energy saving:

$$N_{opt} = [(R \times U \times \alpha) / (C \times \eta \times (1 - P_{failure}))] + N_{redundancy} + N_{sleeping}$$

α is workload complexity factor (1.0-2.5), and $N_{sleeping}$ refers to the optimal sleeping node pool size which is decided by predicting the load and matching with the needs of energy saving [21].

B. Intelligent Sleeping Node Activation Algorithms

A strategy with several limits is used by the system which activate the sleeping nodes: (1) Instant Activation when current Load > 85% the sleeping node is turned on within 30 seconds, (2) Predictive Activation when predicted Load > 70% nodes are planned to turn on based on machine learning predictions, (3) Controlled reduction of scaling when load < 30% for > 10 minutes extra nodes change to sleeping mode for less energy use [22]. The algorithm focuses on past load patterns, analyzes data based on time, and makes real-time predictions to make the best use of resources while making sure that QoS requirements are still being maintained.

IV. EXPERIMENTAL EVALUATION

A. Comprehensive Testbed Setup

The experimental evaluation can be carried out through the testbed. There are 200 fog clusters in the testbed which are spread in different areas. The hardware consists of several levels including Master-of-Masters (Intel Xeon E5-2690 v4, 64GB RAM, 10 Gbps networking), Regional Masters (Intel Xeon E5-2660 v3, 32GB RAM), Cluster Masters (AMD EPYC 7302P, 16GB RAM), Fog Nodes (Raspberry Pi 4, 8GB RAM), and IoT Devices (ESP32- DevKitC). For the software we used Go 1.19 with etcd to manage the agreement between nodes, Python 3.9 with Redis were used for integration, Node.js 16 was used for processing data in fog nodes,

and custom C++ programs were used to make realistic IoT loads. The testbed established categories of services such as smart city monitoring, industrial automation, and mobile IoT applications [23].

TABLE II: PERFORMANCE COMPARISON WITH EXISTING APPROACHES

Architecture	MaxDevices	Latency	Availability	Energy	Failover
Traditional Cloud	100,000	270ms	97.5%	N/A	>5min
Single-Cluster Fog	50,000	145ms	94.2%	12%	45-90s
Hierarchical Fog[17]	250,000	112ms	96.8%	18%	15-30s
HFMCA (Proposed)	1,000,000+	89ms	99.95%	38%	2.3s

V. CONCLUSION

The HFMCA acts as a provider of multiple solutions to a variety of issues associated with fog computing, including but not limited to mastering the master-of-master hierarchies, creating a correctly managed sleeping node, and attaching nodes to the ability to adjust their functions. Some adjustments to the system include: (1) removal of the single points of failure via the use of master-slave redundancy; [2] support of over 1,000,000 IoT devices with a 99.95% uptime via fault-tolerant capability; (3) 67% reduction in latency compared to current cloud-based systems while maintaining the same quality of service; (4) 45% greater efficiency in resource utilization through adaptive scaling and predictive workload adjustments; (5) 38% reduction in energy consumption via activation of sleeping nodes based on predictions[25]; (6) recovery times from failure of less than 3 seconds at any level of hierarchy. Smart cities, industrial IoT and solutions that are optimized for efficiency are the major focus of our solution architecture. Future system improvements will likely focus on AI-based resource optimization, connectivity with the 5G network through the flexible allocation of bandwidth, and the use of quantum-safe methods for securing devices in IoT environments over extended periods[26].

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